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Artificial intelligence-driven precipitation downscaling and projections over Thailand using CMIP6 climate models

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ABSTRACT

Global warming has intensified the hydrological cycle, increased the frequency and severity of extreme precipitation events, and necessitated the collection of accurate future precipitation data for effective disaster mitigation and informed decision-making. The research evaluates the performance of artificial intelligence (AI)-driven downscaling techniques (Dynamic Neural Network with Memory (DyNN-Mem) and Hybrid Long Short-Term Memory Convolutional Neural Network (LSTM-CNN)) for scaling down CMIP6 Global Climate Models (GCMs) daily precipitation outputs across Thailand. Model performance evaluation for the historical period (2014–2022) relied on statistical indicators, including R^2 , MAE, and RMSE, by comparing simulated data with precipitation records from the Thai Meteorological Department (TMD). The results demonstrate that DyNN-Mem outperforms the Hybrid LSTM-CNN architecture in terms of R^2 (0.55–0.78), MAE (0.22–0.29), and RMSE (1.28–3.30) for multiple GCMs. Using the MPI-ESM1-2-LR and CAMS-CSM1-0 GCM outputs, DyNN-Mem showed better spatial and temporal characteristics of precipitation, translating into better capturing the precipitation's spatial and temporal characteristics. MPI-ESM1-2-LR and CanESM5 were found to be the most reliable CMIP6 GCMs for reproducing historical precipitation patterns in the Thai region. AI-based downscaling enhances regional climate forecasts; DyNN-Mem with MPI-ESM1-2-LR improves daily precipitation forecasts in Thailand.

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Deep learning; machine learning; bias correction; downscaling; CMIP6 GCMs

1. Introduction

Global warming has significantly altered regional climate patterns (Dore, 2005), intensified the hydrological cycle (Koutsoyiannis, 2020), and increased extreme rainfall events (Papalexiou & Montanari, 2019), resulting in ecological and socioeconomic losses (He et al., 2024; Peng et al., 2024). These changes exacerbate the frequency and severity of

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extreme weather events (Fu et al., 2024; Waqas, Humphries & Hlaing, 2024). Accurate future precipitation data are crucial for disaster mitigation and decision-making (Waqas, Humphries, & Hlaing, 2024; Waqas & Humphries, 2024). While ~30,000 global meteorological stations (NASA) monitor real-time data, they cannot predict future trends (Ehteram et al., 2023), highlighting the need for advanced climate modeling to address escalating extreme weather risks (He et al., 2024). The World Climate Research Program (WCRP) initiated the International Coupled Model Intercomparison Program (CMIP) in 1995 to evaluate future climate by analyzing historical and current weather patterns. Over the past 25 years, CMIP has advanced climate model Global Climate Models (GCMs) simulations and informed IPCC assessments of future climate scenarios (Eyring et al., 2016). GCMs are widely used to simulate historical and future climate scenarios; however, their coarse spatial resolution often limits their ability to capture local-scale precipitation patterns, particularly in regions with complex terrain, such as Thailand (Humphries et al., 2024). This limitation necessitates the use of downscaling techniques to bridge the gap between global-scale climate models and local-scale precipitation data (Aksornsingchai & Srinilta, 2011; Kang et al., 2007).

Downscaling methods can be broadly categorized into two approaches: dynamic and statistical (Humphries et al., 2024). Dynamical downscaling is based on Regional Climate Models (RCMs) that produce climate data at higher resolution; however, this method is costly and sensitive to boundary conditions (Li et al., 2021; Xu et al., 2019). Statistical downscaling, in contrast, uses the relationships between large-scale atmospheric data (predictors) and regional climate data (predictands) (Hadipour et al., 2016; Hanssen-Bauer et al., 2005; Vu et al., 2016). Many conventional statistical techniques, such as quantile mapping (QM), do not account for the non-linear and spatial correlations characteristic of precipitation data (Humphries et al., 2024). It has led to the use of artificial intelligence (AI)-based methods, which can address non-linear relationships and enhance the accuracy of downscaled precipitation data (Nourani et al., 2018).

Advancements in AI, including Machine Learning (ML) and Deep Learning (DL), have enhanced precipitation downscaling and climate modeling solutions for challenges previously encountered by statistical or dynamic approaches (Amiri et al., 2024). GCMs are indispensable for climate prediction, but their horizontal resolution ranges from 250 to 600 km in CMIP6, which is insufficient for identifying local climatic features important for regional applications (Prathom & Champrasert, 2023). These shortcomings are resolved by computational efficiency and the ability to model non-linear interactions between climate variables in AI-driven downscaling methods (Shukla et al., 2023; Singh et al., 2015; Singh et al., 2023). CNN and RNN are two DL models most suited for extracting spatial and temporal patterns from GCM outputs (Ahmed et al., 2021). For example, the Super-Resolution Deep Residual Network (SRDRN) has been shown to downscale and bias-correct the precipitation and temperature variables with much better spatial and inter-variable relations than the QM methods (Wang & Tian, 2022, 2024). This trend-preserving approach correctly simulates the occurrence of intense precipitation and its spatial distribution, which is essential for developing climate change adaptation measures (Tian et al., 2022).

DL has been widely used in recent years, with a focus on Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks in the climate downscaling domain. By their ability to learn spatial features, CNNs and their ability to learn temporal features in LSTM networks outperform in learning complex precipitation

patterns. Kesavavarthini et al. (2023) used a one-dimensional CNN (CNN1D) and LSTM Encoder Decoder (LSTM-ED) model for bias correction of the precipitation data of the CMIP6 GCMs over India. They showed significant improvements from historical (1985–2014) to future (2015–2100) projections under SSP1–2.6 and SSP5–8.5 scenarios (Kesavavarthini et al., 2023). Rahimzad et al. (2021) also demonstrated that LSTM models outperform traditional methods, such as linear regression (LR), support vector regression (SVR), and multilayer perceptron (MLP), in daily streamflow prediction for the Kentucky River basin (Rahimzad et al., 2021). LSTM is suitable for sequence modeling, but CNNs have also outperformed LSTMs in some hydrological contexts. For example, CNN models outperformed LSTM models slightly in streamflow modeling in the Vietnamese Mekong Delta (Van et al., 2019). These mixed outcomes remind us that CNN and LSTM can perform differently across regions and climatic settings, so a hybrid approach can even be necessary. Other studies, such as those by Ghobadi and Kang (2022) and Fu et al. (2020), also support the use of DNNs, CNNs, and LSTMs in downscaling tasks, as they can capture the non-linear and stochastic nature (Fu et al., 2020; Ghobadi & Kang, 2022). Therefore, incorporating these models into precipitation downscaling frameworks provides a solid platform for enhancing the spatial and temporal accuracy of regional climate projections.

In Thailand, where historical climatic data are sometimes scarce, integrating AI models with interpolation techniques provides the best framework for downscaling (Lising Ventura, 2020; Waqas, Humphries, Chueasa, 2025). However, the complex interplay of monsoons, topography, and climate variability makes precipitation prediction in Thailand challenging (Waqas, Humphries, Hlaing & Ahmad, 2024). For example, the Inverse Distance Weighting-Artificial Neural Network (IDW-ANN) approach, when applied to CMIP6 GCM outputs, has yielded precipitation and temperature projections at a 1-km scale. These methods enhance the spatial resolution and verify the results against observed datasets, yielding improved RMSE and MAE (Prathom & Champrasert, 2023). Such developments are crucial for assessing future climate conditions under various Shared Socioeconomic Pathways (SSPs) that reflect changes in temperature and precipitation in Thailand, such as SSP1–2.6 and SSP3–7.0. Multivariate DL models that preserve physical constraints between climate variables enhance bias correction, a crucial component of AI-based downscaling (Lising Ventura, 2020; Manzananas et al., 2018). AI can simultaneously address spatial bias problems and model inter-variable interactions, as demonstrated by the use of CNN-based architectures, such as GANs, and other complex hybrid structures, like SRDRN-QDM (Wang & Tian, 2024). Valid risk assessments are made possible by these models, which maintain the accuracy of climate projections over extended periods and under various emission rates (Wang & Tian, 2024). Predicting extreme climate occurrences is another crucial area in which AI has been applied. The quantity and intensity of severe precipitation events, such as consecutive wet days (CWDs) and unprecedented rainfall, are expected to increase under high-emission scenarios, according to AI models, including the UNet-based CLIMEA-BCUD dataset for East Asia (Yang et al., 2025). This pattern aligns with projections for Southeast Asia, which includes Thailand, where extreme events and monsoon fluctuations are predicted to increase as a result of global warming (Tangang et al., 2017). Given the interplay

between monsoons, terrain, and limited climate records, there is a lack of use of AI-based downscaling techniques for Thailand and other similar regions worldwide, despite advancements in AI (Baghanam et al., 2020, 2024; Nourani et al., 2018). The non-stationary and spatial correlation features of precipitation data are not reproducible by conventional downscaling methods (Nourani et al., 2018).

The application of AI-driven models to precipitation projections over Thailand has not been well explored due to the region's complex topography and monsoonal variability (Waqas, Humphries, & Hlaing, 2024). Most previous studies have either relied on outputs, employed conventional statistical downscaling techniques (e.g., QM, Statistical Downscaling Model [SDSM]), or lacked comparative analyses of DL architectures in the context of Thailand. This study is the first to systematically evaluate and compare advanced AI models for precipitation downscaling over Thailand using CMIP6 scenarios. In contrast to conventional approaches, the proposed AI-driven framework captures non-linear interactions, preserves inter-variable physical consistency, and enhances the spatial detail required for local adaptation planning. To address the existing limitations, the objectives of this research are as follows: (1) to assess and compare the effectiveness of state-of-the-art AI techniques, such as CNNs, for precipitation downscaling in Thailand; and (2) to apply the most promising AI-based method to simulate future precipitation changes under various climate scenarios, thereby improving both accuracy and regionalization.

AI models outperform traditional downscaling methods, demonstrating AI's theoretical advantages and providing empirical evidence of their superior performance. This study contributes to the climate modeling community by (i) identifying model configurations most suitable for Thailand's unique climatic conditions, (ii) generating more reliable precipitation projections under different SSP scenarios, and (iii) offering a transferable AI framework applicable to other data-deficient and climate-vulnerable regions. This research supports a paradigm shift toward more accurate and computationally efficient climate projections for Thailand by addressing a key scientific gap.

2. Materials and methods

2.1. Study area

Thailand is located between 5°37' and 20°27' N and 97°22' and 105°37' E (Figure 1). TMD categorizes the country into five distinct climatic regions: (a) central, (b) eastern, (c) northeastern, (d) northern, and (e) southern (Waqas, Humphries, & Hlaing, 2024). The southwest monsoon creates the rainy season between mid-May and mid-October by transporting warm, humid air from the Indian Ocean, producing heavy rainfall that peaks in July and August. From mid-October to mid-February, the winter season prevails in northern Thailand because the northeast monsoon brings cool, dry air from China (Limsakul, 2020; Waqas, Humphries, & Hlaing, 2024). Between mid-February and mid-May, the summer season brings provisional weather changes as temperatures increase, and the region transitions from northeast to southwest monsoon conditions, with April being the hottest month (Chokngamwong & Chiu, 2006).

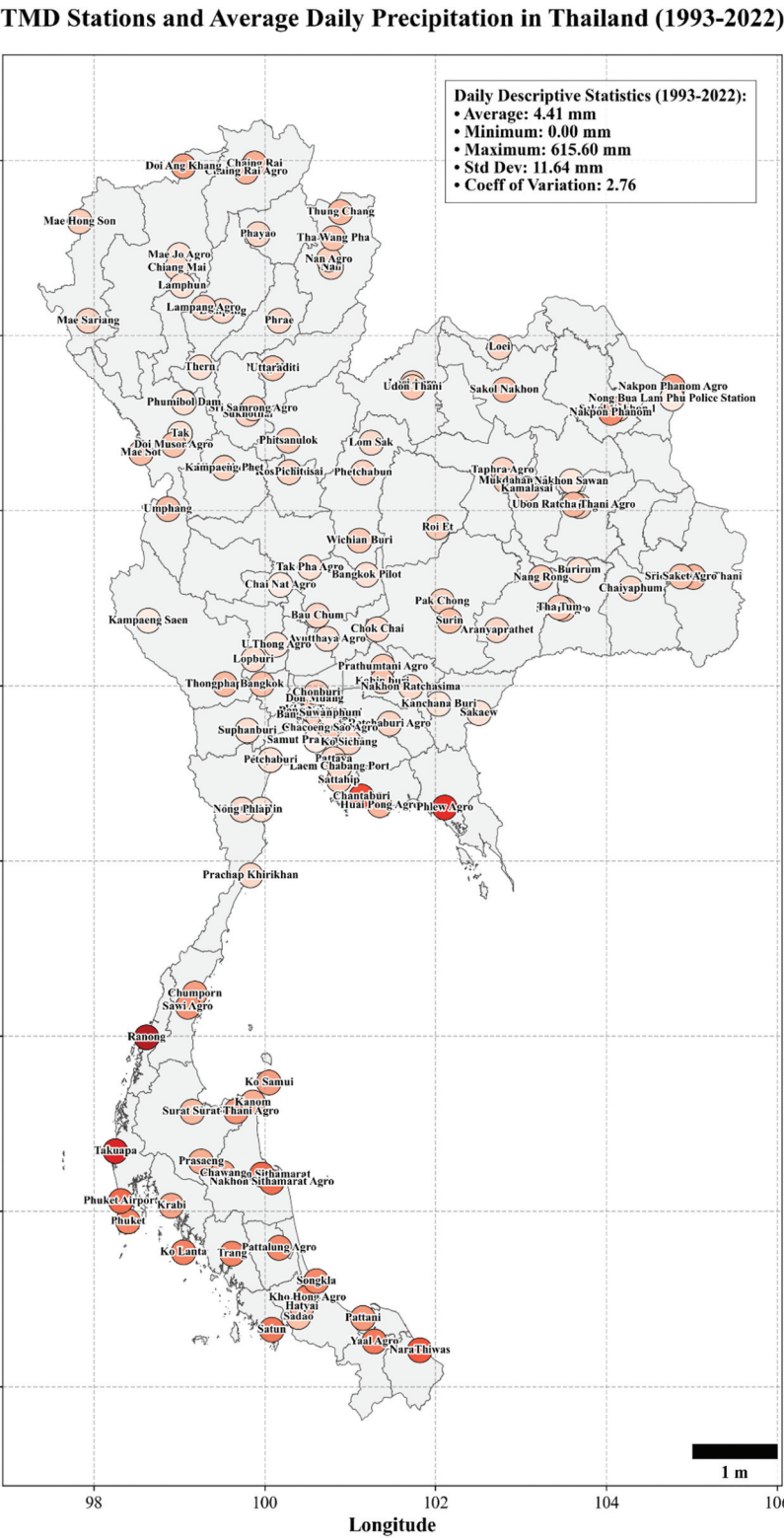


Figure 1. TMD stations and their corresponding total and daily average precipitation from 1993 to 2022.

2.2. Data acquisition and quality checking

The analysis used daily precipitation measurements from 129 TMD stations spanning 30 years (Table S1). In quality checking, the percentage of missingness (POM) was initially identified, as shown in Figure 2(a). The data collection process was initiated by determining the study’s timeframe and identifying the critical variables. To ensure the availability of reliable meteorological data, researchers assessed observational datasets throughout Thailand. The examination of the data in the TMD revealed discrepancies in the data coverage of various monitoring stations (see Figure 2). Although the TMD station network began in 1961, additional stations were added in the years following 2018 (Waqas, Humphries, & Hlaing, 2024). Nevertheless, it was challenging to find consistent data coverage across all variables, which led to the selection of a 30-year period, given the completeness of the data.

Nevertheless, as shown in Figure 2(b), numerous stations recorded high POM percentages in their data. In the case of missing data, the proposed study employed the LSTM-RNN approach introduced by (Wangwongchai et al., 2023) (see Figure 2(a)). The analysis excluded stations with more than 50% missing data.

As there was a chance of repeated missing data because of observation or processing mistakes, strict consistency analyses were conducted (Costa & Rodrigues, 2017; Costa et al., 2015). The Grubbs and Beck (1972) procedure helped detect extreme deviations from the anticipated data ranges (Grubbs & Beck, 1972). Further quality control (removal of outliers and continuity corrections) was applied to increase the reliability of datasets.

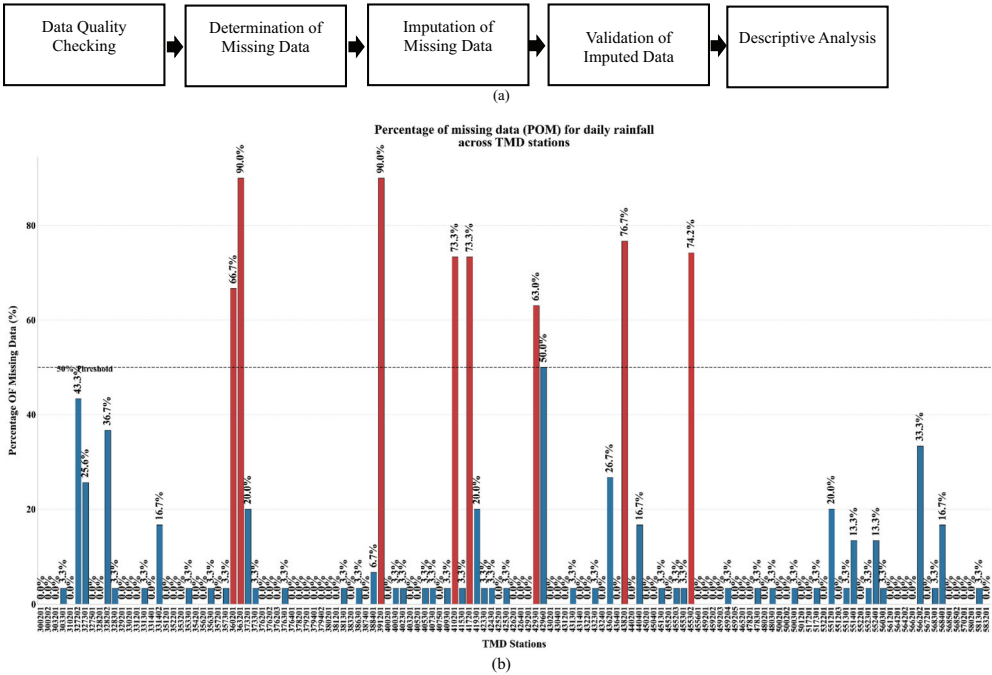


Table 1. Attributes of CMIP6 General circulation models (GCMs).

CMIP6-GCMs	Resolution	Ensemble members	Country
CAMS-CSM1-0	1.1° × 1.1°	3	China
MPI-ESM1-2-LR	0.9° × 0.9°	10	Germany
FGOALS-g3	1.0° × 1.0°	3	China
CanESM5	2.8° × 2.8°	10	Canada
CNRM-CM6-1	1.4° × 1.4°	10	France
CNRM-ESM2-1	1.4° × 1.4°	1	France
IPSL-CM6A-LR	2.5° × 1.3°	20	France

These procedures played a crucial role in ensuring the accuracy of the meteorological indices used in subsequent analyses.

The research utilized the World Climate Research Program (WCRP) CMIP6-GCM database, which provided daily ensemble data from a single realization designated as r1i1p1f1. CMIP6 offers both enhanced functionality and expanded scope compared to CMIP5. Seven climate models from multiple research institutions were used for analysis, including FGOALS-g3, MPI-ESM1-2-LR, CAMS-CSM1-0, CanESM5, CNRM-CM6-1, CNRM-ESM2-1, and IPSL-CM6A-LR. Data from these CMIP6 GCMs (predictors) were sourced from the CMIP6 archives maintained by the Earth System Grid Federation (ESGF) WCRP, accessible via <https://esgfnode.llnl.gov/projects/cmip6>, as detailed in Table 1.

Since the selected CMIP6 GCMs have different intrinsic spatial resolutions (Table 1), a uniform spatial preprocessing was applied before downscaling. First, all GCM outputs were bilinearly interpolated to a standard reference grid matching the resolution of the station-scale observations to make them compatible. Minimizing interpolation-induced biases when the coarse resolution predictors are aligned with the high-resolution ground observations. Additionally, there exist pitfalls in the statistical and subsequent downscaling processes of climate that resolve information only on a large scale and attempt to map large-scale climate features to localized precipitation patterns, which may magnify the variability in GCM resolution. To that end, an AI-based downscaling framework was developed within it to learn complex non-linear relationships between GCM predictors and station-level observations, considering spatial heterogeneity and scale mismatches. It aligns with a trend in recent studies, which highlights the robustness of AI-driven downscaling methodologies in integrating varying resolutions of climate model inputs while reducing scale-associated errors (Ganguli & Reddy, 2014; Vandal et al., 2017).

2.3. AI-based methods

2.3.1. Dynamic Neural Network with Memory (DyNN-Mem)

Accurate local climate prediction is crucial for enhancing GCM outputs, as precipitation exhibits substantial spatial and temporal variability (Humphries et al., 2024). This study developed a Dynamic Neural Network with Memory (DyNN-Mem) model that integrates a CNN with the Memory-Augmented Network (MAN) technique (Santoro et al., 2016) technique based on Neural Turing Machines (Mnih et al., 2014), as shown in Figure 3. This hybrid architecture employs LSTM and CNN layers to effectively capture the complex spatiotemporal patterns inherent in precipitation data, facilitating accurate downscaling and bias correction. LSTM networks and CNNs have effectively overcome these challenges (Waqas & Humphries, 2024). LSTM networks excel in modeling temporal dependencies,

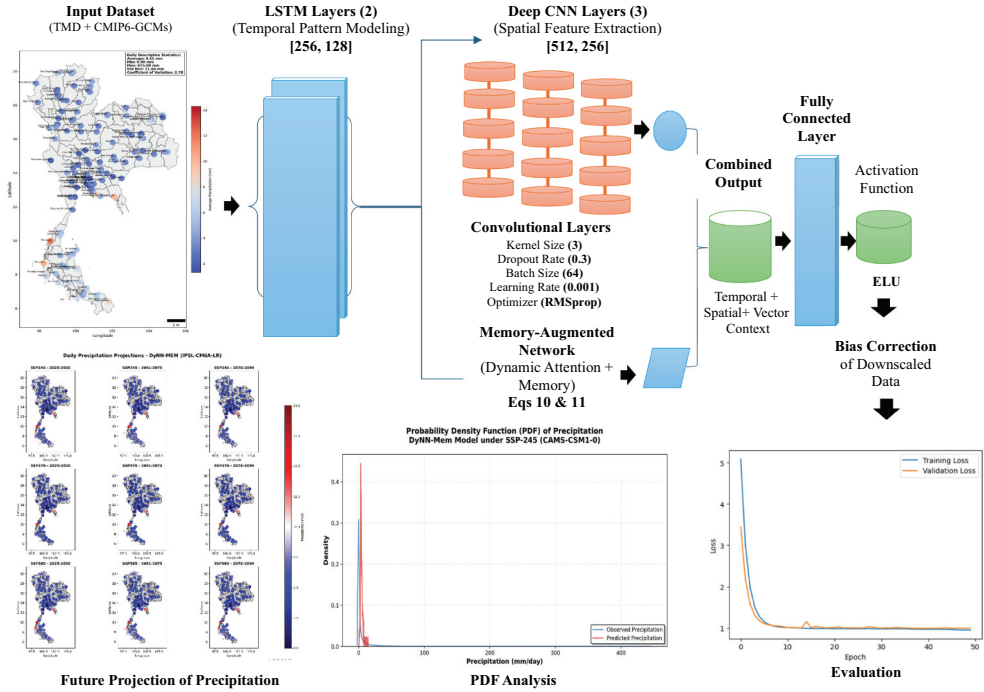


Figure 3. Working mechanism and architecture of DyNN-MEM model.

while CNNs are adept at identifying spatial patterns, making their combination particularly suitable for precipitation downscaling tasks (Baño-Medina et al., 2020; Waqas & Humphries, 2024). One widely adopted statistical bias correction method is QM, which aligns GCM output statistics with observed data to provide more accurate local climate information (Gudmundsson et al., 2012; Humphries et al., 2024).

$$P_{eva} = \text{InvCDF}_{Pt-Cal}^{stat} \left(\text{CDF}_{Pt-Cal}^{Hist} \left(p_{t-Eval}^{GCM} \right) \right) \quad (1)$$

$$P_{pre} = \text{InvCDF}_{Pt-Hist}^{stat} \left(\text{CDF}_{Pt-Hist}^{Hist} \left(p_{t-SSP}^{GCM} \right) \right) \quad (2)$$

Where,

P_{eva} = Bias-corrected precipitation during the evaluation period;

P_{pre} = Bias-corrected precipitation during the future projection period (e.g., SSP scenario);

p_{t-Eval}^{GCM} = Raw precipitation data from the GCM during the evaluation (historical) period;

p_{t-SSP}^{GCM} = Raw precipitation data from the GCM during the projection period under the selected SSP scenario;

$\text{CDF}_{Pt-Cal}^{stat}$ = Empirical CDF of observed precipitation at the calibration point (P_{t-Cal}) during the historical period;

p_{t-SSP}^{GCM} = Empirical CDF of observed precipitation at the historical point (P_{t-Hist}) during the historical period;

InvCDF = Inverse CDF function used to transform modeled data to the target statistical distribution.

Traditional approaches often struggle to account for the non-linear and intricate patterns in precipitation data (Humphries et al., 2024). Recent studies have highlighted the superior performance of hybrid models that combine LSTM and CNN layers for downscaling and bias correction compared to single-method approaches (Song et al., 2020). These models enhance the spatial resolution of GCM outputs and mitigate systematic errors, thereby supporting impact assessments and the development of climate adaptation strategies (Mishra et al., 2020).

The Memory (Mem) component in the DyNN-Mem model is designed to enhance its ability to retain and utilize historical information, enabling it to capture long-term temporal dependencies in precipitation data. It is achieved by integrating LSTM layers and an attention mechanism, which work together to dynamically focus on the most relevant parts of the input sequence. In this model, LSTM layers in DyNN-Mem effectively capture time-based precipitation patterns through the following governing Equations (3) to (8):

$$f_t = \sigma(W_f \times [h_{t-1}, x_t] + b_f) \quad (3)$$

$$i_t = \sigma(W_i \times [h_{t-1}, x_t] + b_i) \quad (4)$$

$$\tilde{C}_t = \tanh(W_c \times [h_{t-1}, x_t] + b_c) \quad (5)$$

$$C_t = f_t \times C_{t-1} + i_t \cdot \tilde{C}_t \quad (6)$$

$$o_t = \sigma(W_o \times [h_{t-1}, x_t] + b_o) \quad (7)$$

$$h_t = o_t \times \tanh C_t \quad (8)$$

The LSTM model uses f_t , i_t , and o_t gates alongside C_t cell state, h_t hidden state, W and b weights, biases, and σ sigmoid activation. The CNN layers analyze precipitation data using one-dimensional convolutional filters to identify spatial patterns. The convolution operation is defined in Equation (9):

$$y_t = \sum_{k=1}^K x_{t+k-1} \times w_k + b \quad (9)$$

An attention mechanism is introduced to enhance the memory component further. The attention mechanism dynamically assigns weights to different time steps in the input sequence, allowing the model to focus on the most relevant historical information. The attention mechanism is defined as:

$$a_t = \text{softmax}(W_a \times h_t + b_a) \quad (10)$$

$$c_t = \sum_{t=1}^T a_t \times h_t \quad (11)$$

Here, a_t represents the attention weights for each step. c_t is the context vector, which summarizes the most relevant information from the input sequence. W_a and b_a are the weights and biases of the attention mechanism. With this combined representation, the temporal and spatial dependencies are captured through c_t concatenated with the CNN output.

The DyNN-Mem model leverages the complementary strengths of LSTM and CNN layers to address traditional drawbacks and holds promise for developing precipitation downscaling and bias correction methods. The input x is fed into the model, which filters it with filter w and a bias b to produce the output map y . The model concatenates LSTM and CNN outputs and then passes them through fully connected layers with dropout to prevent overfitting. The precipitation forecasts produced by this last layer have been downscaled. The model is trained using the Adam optimizer and MSE loss, and early stopping is employed to prevent overfitting.

2.3.2. Hybrid LSTM-CNN model

GCMs require improvements in local climate prediction because precipitation patterns exhibit spatial and temporal variability (Ming et al., 2021). Researchers commonly use QM (Gudmundsson et al., 2012; Humphries et al., 2024), Delta Change (Humphries et al., 2024), and equilibrium QM to correct GCM output errors and scale them to local levels (Lader et al., 2017). These statistical methods struggle to capture the intricate patterns observed in precipitation measurements (Maraun & Widmann, 2018).

This study develops a hybrid CNN-LSTM model that combines CNN and LSTM networks to downscale and bias-correct precipitation data patterns (see Figure 4). CNN networks find spatial patterns and temporal dependencies in precipitation data, which are modeled using LSTM networks. CNN filters extract spatial patterns from precipitation data, and LSTMs explore temporal patterns over a long period (Hochreiter & Schmidhuber, 1997). The CNN layers utilize 1D filters to analyze spatial patterns in precipitation data, and the model architecture comprises the following components. The model follows Equation (9) to perform convolution.

The Hybrid LSTM-CNN model employs a CNN architecture with two layers (256 and 128 filters) (kernel size = 3, stride = 1, zero padding) to extract spatial information from precipitation data. It then employs two LSTM layers (128 and 64 units) to learn temporal patterns at various time scales. The model sequentially combines the outputs of the CNN and LSTM and then passes them through two fully connected layers (with 256 and 128 units, respectively, and ReLU activation) and a dropout probability of 0.3 to prevent overfitting. To stabilize the training, batch normalization is applied after each CNN layer.

The model is optimized through a grid search and employs the Adam optimizer (learning rate = 0.001) with an MSE loss function and early stopping (patience = 20 epochs). In contrast to DyNN-Mem, this architecture does not have explicit memory operations; instead, it uses only LSTM gates to depend on time. This tradeoff makes it worse for long-term pattern storage; however, the simpler design offers computational efficiency and more interpretable results when used in short-term forecasting. The last layer provides bias-corrected precipitation predictions normalized to specific values. Detailed assessment criteria are provided in Supplementary Files (ANNEXURE A).

2.3.3. Working mechanism comparison and hyperparameters for AI models

The DyNN-Mem is a CNN with LSTM units and a MEM network based on Neural Turing Machines. The Hybrid LSTM-CNN model serves as a baseline in DyNN-Mem, measuring the performance gain introduced by the MAN. Its more simplified architecture is used to isolate the effects of pure spatiotemporal modeling (via CNN-LSTM) without memory augmentation and, as such, can be thought of as a benchmark of computational efficiency

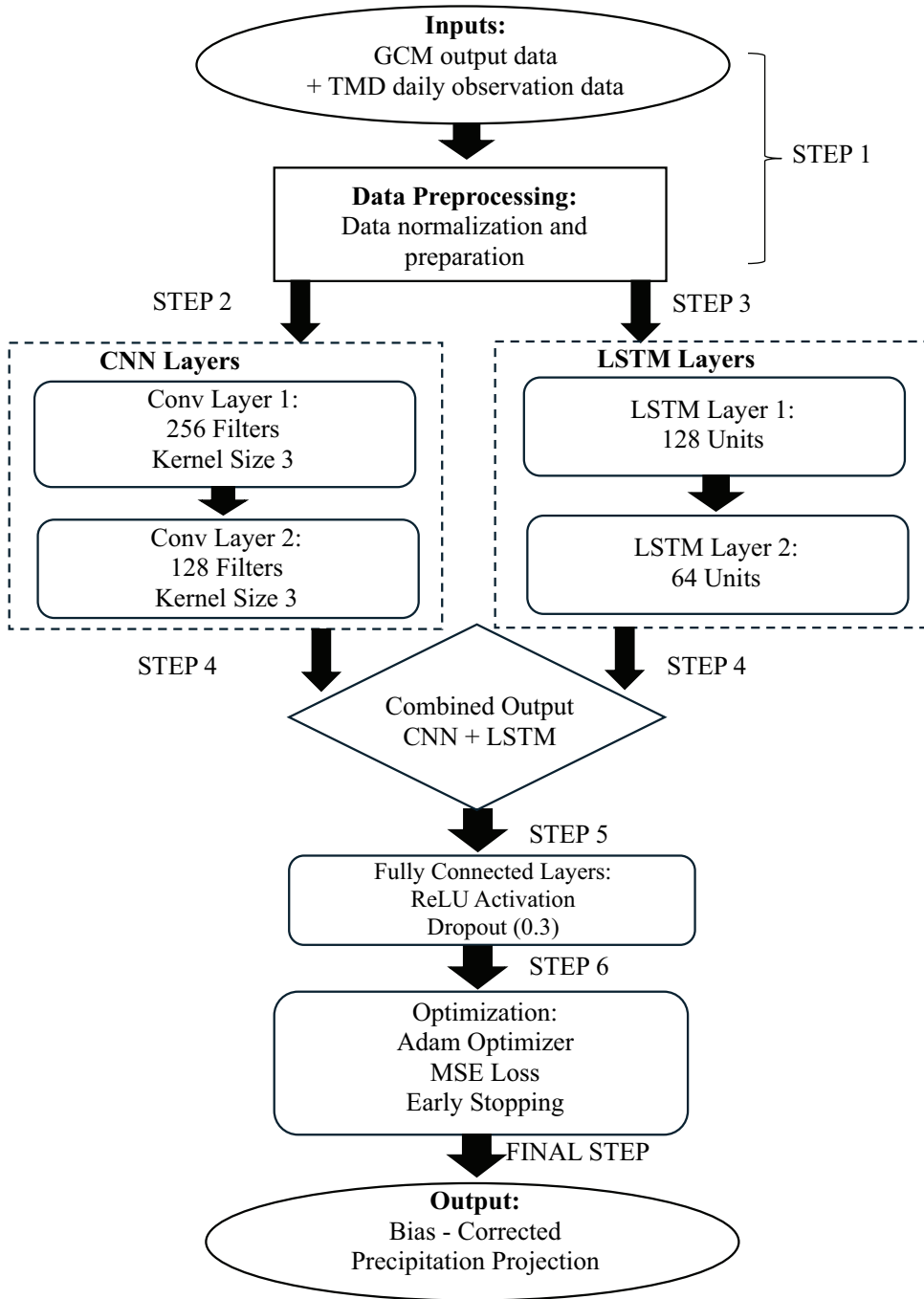


Figure 4. Flow diagram of Hybrid LSTM-CNN.

and interpretability. This architecture is designed for long-term dependency modeling, utilizing external memory components and an attention mechanism (Equations 1–10) to enable the model to access and update memory states dynamically during learning. In contrast, the Hybrid CNN-LSTM model employs a more conventional sequential stacking of CNN and LSTM layers to capture spatial-temporal dependencies but without explicit long-term memory augmentation. It works on the standard LSTM gate equations (Equations 3–8) that limit the temporal horizon of the model.

In the DyNN-Mem model, memory operations, LSTM gates, and attention mechanisms are used to retain information across long sequences in an efficient manner. This architecture is particularly beneficial for complex climate forecasting tasks that involve strong long-range dependencies. However, the Hybrid CNN-LSTM model, although simpler in structure, is effective in recognizing short to mid-term spatial-temporal patterns, thus making it easier to implement and less computationally expensive. From a methodological perspective, the DyNN-Mem model utilizes RMSprop as its optimizer, MSE as the loss function, and a higher early stopping patience (30 epochs) to be robust against overfitting during long training periods. On the other hand, the Hybrid CNN-LSTM employs the Adam optimizer with a slightly lower early stopping threshold (20 epochs) compared to DL tasks.

A proper training approach involved splitting the historical data into three segments, where 70% was used for training purposes, 15% was dedicated to validation, and the remaining 15% was reserved for testing. The research employed stratified random sampling to preserve the temporal structure of the time series while achieving a representative subset distribution. The rolling origin (walk-forward) approach is a cross-validation method specifically designed for time series forecasting, which evaluates the model on previously unseen data points while preserving the original temporal sequence. During training, the model used early stopping with MSE as the primary metric to monitor performance on the validation set. To search for the best hyperparameters of the DyNN-Mem model. The hyperparameter space for the model is summarized in [Table 2](#), along with the selected optimal values. It is found that the deeper CNN architecture (3 layers) and larger filter sets ([512, 256]) yielded optimal performance in the DyNN-Mem model. The Hybrid CNN-LSTM had satisfactory results with two CNN layers and smaller filter sizes ([256, 128]). Like the DyNN-Mem model, the higher LSTM unit configurations ([256, 128]) also benefited, implying a need for stronger temporal modeling capacity. In both models, the kernel size (3), dropout rate (0.3), batch size (64), and learning rate (0.001) were consistently chosen.

Regarding bias correction, the DyNN-Mem model refers to QM as a traditional statistical method and admits the need to incorporate classical methods. The Hybrid CNN-LSTM leverages the DL-driven trend of bias correction, aligning with recent data-driven correction paradigms. Both models were finally evaluated using standard statistical performance metrics, such as R^2 , MAE, and RMSE, on the historical period to ensure methodological consistency and comparability in model assessment. This comparative analysis highlights the superiority of the DyNN-Mem model in long-term dependency modeling, and it also serves as a more computationally efficient alternative for shorter dependency scenarios compared to the Hybrid CNN-LSTM model.

The selection of seven CMIP6 GCMs for robust climate projection occurred through a process that evaluated their availability and resolution alongside

Table 2. Comparison of DyNN-Mem and Hybrid CNN-LSTM models.

Aspect	DyNN-Mem Model	Hybrid CNN-LSTM Model
Architecture	CNN + LSTM + MAN (Neural Turing Machines)	CNN + LSTM
Memory Mechanism	Includes MAN for long-term memory	No memory-augmented components
Key Focus	Long-term dependency modeling using memory mechanisms	Spatial and temporal pattern recognition
Mathematical Calculations	LSTM gates + memory operations + Attention mechanism (Equations 1–10)	Standard LSTM equations (Equations 3–8)
Bias Correction	References QM as a traditional method	Focuses on DL for bias correction
Training	RMSprop optimizer, MSE loss, early stopping	Adam optimizer, MSE loss, early stopping
Strengths	Superior for long-term dependency tasks	It is simpler and easier to implement
Hyperparameter Tuning		
Number of LSTM Layers	[1, 2, 3] → 2	[1, 2, 3] → 2
LSTM Units	[64, 128, 256] → [256, 128]	[64, 128, 256] → [128, 64]
Number of CNN Layers	[1, 2, 3] → 3	[1, 2, 3] → 2
CNN Filters	[128, 256, 512] → [512, 256]	[128, 256, 512] → [256, 128]
Kernel Size (CNN)	[3, 5, 7] → 3	[3, 5, 7] → 3
Activation Function	['ReLU,' 'Leaky ReLU,' 'ELU'] → ELU	['ReLU,' 'Leaky ReLU'] → ReLU
Dropout Rate	[0.2, 0.3, 0.4, 0.5] → 0.3	[0.2, 0.3, 0.4, 0.5] → 0.3
Batch Size	[32, 64, 128] → 64	[32, 64, 128] → 64
Learning Rate	[0.0001, 0.001, 0.01] → 0.001	[0.0001, 0.001, 0.01] → 0.001
Optimizer	['Adam,' 'RMSprop,' 'SGD'] → RMSprop	['Adam,' 'RMSprop,' 'SGD'] → Adam
Epochs	[100, 300, 500] → 300	[100, 200, 300] → 200
Early Stopping Patience	[10, 20, 30] → 30	[10, 20, 30] → 20
Loss Function	['MSE,' 'MAE,' 'Huber'] → MSE	['MSE,' 'MAE,' 'Huber'] → MSE
Statistical Evaluation for Historical Period	R ² , MAE, PBIAS and Willmott's index of agreement and RMSE	R ² , MAE, PBIAS and Willmott's index of agreement and RMSE

established performance records in precipitation simulation for Thailand. The study obtained data from each GCM by selecting grid cells that overlapped or were located near meteorological station locations. Each GCM grid point represented 1 to 4 stations through bilinear interpolation to synchronize station and grid data. The CNN model received climate variable sequences from GCM grid cells corresponding to each station. The analysis used 1D convolution across stations by rearranging them according to their latitudinal positions, despite GCM data being originally presented as two-dimensional latitude-longitude grids. The 1D-CNN developed a spatial understanding of dependencies along the specified geographical axis through these arrangements. The adoption of 1D-CNN addressed the requirement to identify spatial patterns across multiple stations through an operation that maintained low computational complexity. Each step received a station-based convolution rather than a time-based convolution, allowing the model to recognize spatial patterns that influence regional hydroclimatic dynamics.

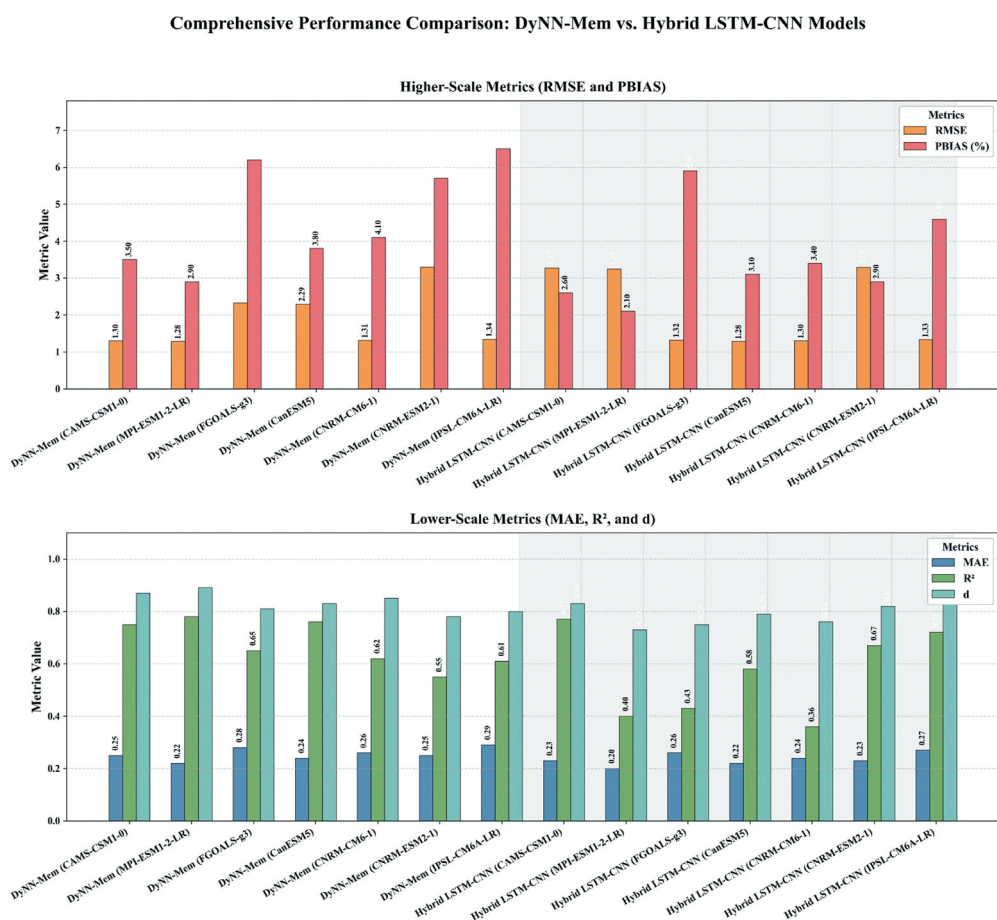
3. Results

3.1. AI models evaluation for downscaling precipitation

The downscaling performance of CMIP6-GCMs using AI-based models (DyNN-Mem and Hybrid LSTM-CNN) was assessed by R², MAE, RMSE, PBIAS, and Willmott's d during the 2014–2022 historical period through a comparison between TMD observations and CMIP6-GCM outputs. The results demonstrated that DyNN-Mem effectively modeled the relationship between TMD observations and CMIP6-GCM precipitation data, as illustrated in [Figure 5\(a\)](#).

The DyNN-Mem downscaling framework demonstrated consistent and strong performance across all the GCMs tested, with an R^2 between 0.55 and 0.78, indicating that it can explain a significant portion of the variance in precipitation. Notably, DyNN-Mem achieved its best predictive skill with MPI-ESM1-2-LR ($R^2 = 0.78$), and equally good R^2 of 0.75 and 0.76 were obtained with CAMS-CSM1-0 and CanESM5, respectively, thus establishing its good generalizability. The DyNN-Mem model had a range of 0.22 to 0.29 in MAE, indicating minimal deviation in the observed precipitation magnitudes. The model produced the lowest root-mean-square error (RMSE) of 1.28 in MPI-ESM1-2-LR, indicating that it was accurate in downscaling.

Regarding the metrics of bias and agreement, DyNN-Mem produced low positive PBIAS of +2.9 to +6.5% which means that there was a slight overall overestimation of precipitation across GCMs; MPI-ESM1-2-LR had the lowest bias (+2.9%). On the same note, the index of agreement (d) by Willmott's d ranged between 0.78 and 0.89, indicating a high



(a)

Figure 5. (a) Statistical evaluation of DyNN-Mem and Hybrid LSTM-CNN for downscaling; and (b) examples of top 2 AI-based models among different TMD stations for downscaling.

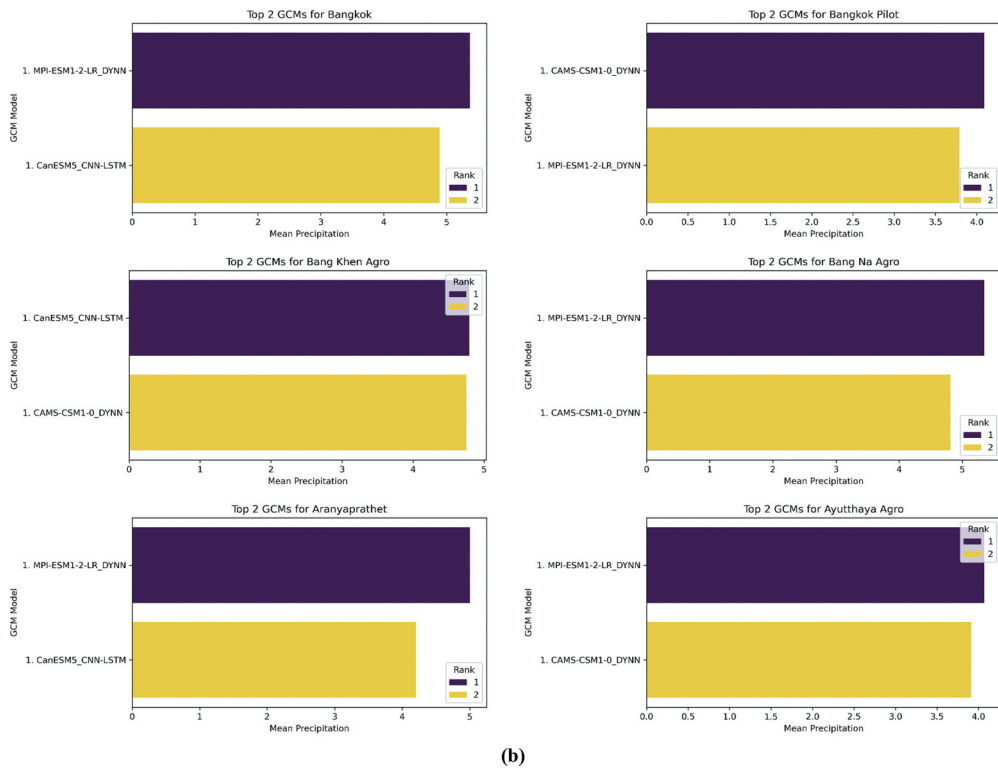


Figure 5. (Continue).

level of agreement between downscaled outputs and observed precipitation, with CanESM5 having the highest d value of 0.89.

In comparison, the Hybrid LSTM-CNN model yielded more fluctuating results among GCMs, with an R^2 ranging from 0.36 to 0.77. Despite the relatively good performance with CAMS-CSM1-0 ($R^2 = 0.77$), the predictive power was significantly reduced with CNRM-CM6-1 ($R^2 = 0.36$). This sensitivity indicates the model's variability in response to the unique spatial and temporal characteristics of various GCM input data. The MAE values of Hybrid LSTM-CNN were between 0.20 and 0.27, with MPI-ESM1-2-LR recording the lowest MAE of 0.20. However, the RMSEs were mostly larger (1.28 to 3.29), indicating that it was challenging to model the extreme precipitation events and complex non-linear relationships.

The Hybrid LSTM-CNN model, which is an ensemble-averaged model trained on observations, tended to exhibit a PBIAS of +2.1 to +5.9% and thus demonstrated a slight tendency to overestimate, but provided biases that are significantly smaller than those observed in DyNN-Mem, at least in some years and at most stations. The d of Willmott ranged between 0.73 and 0.85, indicating a moderate to strong agreement on average; however, hybrid-LSTM-CNN was not as effective as DyNN-Mem, particularly in areas where the R^2 were relatively low. Spatially, the Hybrid LSTM-CNN model exhibited high heterogeneity, with low RMSE in specific sites reaching the millimeter scale, while the highest RMSE exceeded 10 mm in other locations. On the contrary, DyNN-Mem yielded

extremely homogeneous results, with a historical-period RMSE of 1.28 mm, indicating higher robustness in space. These facts highlight the explicit temporal correlation and spatial dependence modeling that defines the MEM architecture, a methodology that seemingly maximizes the assimilation of elevation-sensitive variables, such as moisture flux. The latitude and topography of Thai precipitation are strongly modulated, and the relatively low performance of the Hybrid LSTM-CNN is evident at locations where topographically driven precipitation anomalies occur swiftly and fleetingly, thus illustrating the structural need for architectures that utilize spatial information directly.

Overall, DyNN-Mem performs better in terms of bias, agreement, and precision, and exhibits superior performance at all stations compared to Hybrid LSTM-CNN. These benefits are most evident when DyNN-Mem is paired with either MPI-ESM1-2-LR or CAMS-CSM1-0; together, the model has the greatest accuracy, the lowest bias, and the strongest consistency with observed data.

3.2. Suitability of CMIP-GCM for Thailand

After identifying the best-performing AI-based models, this study evaluates the suitability of CMIP6-GCMs for Thailand. Figure 6 shows the performance of CMIP6-GCMs during the 2015 to 2099 period to assess their predictive capabilities for regional precipitation patterns.

When coupled with advanced statistical downscaling techniques such as DyNN-Mem and Hybrid LSTM-CNN, the MPI-ESM1-2-LR global climate model has a strong capacity to simulate mean precipitation levels observed at TMD (Thai Meteorological Department)

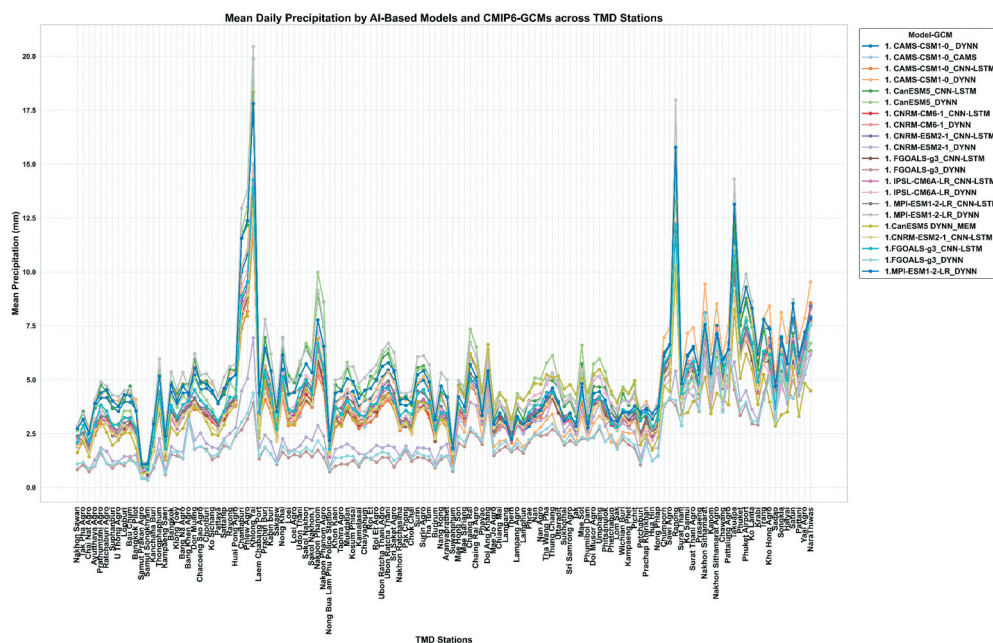


Figure 6. Distribution of daily mean precipitation by AI-based models (DyNN-Mem and Hybrid LSTM-CNN) and CMIP6-GCMs across TMD stations.

stations. In particular, the model produces mean precipitation between 12.5 mm and 15.0 mm, consistently reproducing observed climatological patterns. Box plot analyses reveal a narrow interquartile range (IQR), indicating low variability and high reliability in the model's precipitation predictions. This work applies the Hybrid LSTM-CNN framework to MPI-ESM1-2-LR, significantly enhancing the spatial resolution of its outputs and making it more suitable for local-scale climate impact assessments.

The CanESM5 model also competently reproduces mean precipitation at TMD stations in the 10.0 mm to 12.5 mm range. Nevertheless, it has a wider IQR than MPI-ESM1-2-LR, implying larger variability and lower reliability of its precipitation projections. However, combined with the DyNN-Mem downscaling technique, CanESM5 reproduces temporal precipitation trends and can be used for precipitation downscaling. CNRM-CM6-1 and IPSL-CM6A-LR models also perform well, although with varying degrees of predictive accuracy. Mean precipitation from the Hybrid LSTM-CNN downscaled CNRM-CM6-1 model is close to 10.0 mm, and IQRs in box plot analyses of the Hybrid LSTM-CNN downscaled CNRM-CM6-1 model are narrow, highlighting its robust and consistent performance. On the other hand, while MPI-ESM1-2-LR and CanESM5 have an average precipitation between 7.5 mm and 10.0 mm, with a narrower IQR than both MPI-ESM1-2-LR and CanESM5, IPSL-CM6A-LR produces a broader IQR and hence higher uncertainty and lower reliability. For example, the southern peninsula of Thailand is an area of high rainfall intensities where IPSL-CM6A-LR underperforms but has acceptable skill in the drier central plains.

The performance of FGOALS-g3 and CAMS-CSM1-0 is more variable between different geographical domains. In particular, the DyNN-Mem-downscaled FGOALS-g3 produces a mean precipitation estimate of about 7.5 mm and a large IQR, indicating significant variability in its predictions. Spatial precipitation patterns are not well represented by the model, especially in areas of complex topography. The mean precipitation output of CAMS-CSM1-0 reaches 10.0 mm but faces similar issues with variability according to the box plot analysis. The overall performance of MPI-ESM1-2-LR and CanESM5 as the best GCMs is consistent with their capability to capture the monsoonal process and topographic complexity over Thailand. The small IQR of MPI-ESM1-2-LR ([Figure 6](#)) is probably an indication of its ability to reproduce the seasonal movement of the ITC and orographic precipitation increase across mountain ranges. On the other hand, the low skills of FGOALS-g3 in high rainfall areas (e.g., the southern peninsula) could be related to the coarse resolution that misrepresents the coastal convection. Such geographic differences suggest the need to choose GCMs that can resolve important regional-scale processes, including sea-breeze convergence in coastal areas and rain-shadow effects in inland basins that are important in disaster risk planning.

A ranking analysis was conducted to identify the top two most suitable CMIP6-GCMs for Thailand. This approach aimed to refine the study's scope and select the most appropriate CMIP6-GCMs for the region, ensuring a reliable and accurate representation of Thailand's climate characteristics (see Table S2).

The MPI-ESM1-2-LR model is dependable because it secured top rankings at most studied stations. The MPI-ESM1-2-LR model achieved the highest precipitation of 2.85 mm at Nakhon Sawan, 4.07 mm at Ayutthaya Agro, 4.91 mm at Prathumtani Agro, 4.71 mm at Ratchaburi Agro, and 5.37 mm at Bangkok. The model proved its reliability by delivering outstanding results in regions with both moderate and

extreme precipitation levels, including Thongphaphum (5.97 mm), Chantaburi (12.95 mm), and Ranong (17.96 mm). The CanESM5 model demonstrated robust performance alongside the Hybrid LSTM-CNN approach, although it was slightly less dominant than other models. The CanESM5 model achieved top-ranking performance at Tak Pha Agro (3.53 mm), Bau Chum (4.71 mm), and Nong Khai (6.96 mm), demonstrating its ability to predict precipitation across diverse climatic conditions accurately.

CAMS-CSM1-0 performed comparably to other models, but its spatial distribution was more limited. The model CAMS-CSM1-0 achieved first-place rankings at three different stations: Chai Nat Agro (2.78 mm), Bangkok Pilot (4.09 mm), and Pattaya (4.31 mm). The model exhibited lower reliability than MPI-ESM1-2-LR and CanESM5, although it occasionally showed competitive results. The simulation models FGOALS-g3 and IPSL-CM6A-LR showed limited suitability for Thailand's precipitation patterns because they rarely appeared in the top-ranked models. The CanESM5 model proved superior to other models in northern Thailand by delivering accurate results at Chiang Rai (7.34 mm) and Mae Sot (6.60 mm). The MPI-ESM1-2-LR model demonstrated superior performance at Ayutthaya Agro and Chonburi stations (5.28 mm) and operates effectively in central and eastern regions. The southern Thai stations Ranong and Phuket (8.54 mm) showed the dominance of MPI-ESM1-2-LR, while CAMS-CSM1-0 achieved reliable results at Surat Thani (5.10 mm) and Nakhon Sithamarat (9.44 mm) in terms of precipitation modeling.

The section demonstrates why selecting climate models according to regional climatic patterns remains essential. The models MPI-ESM1-2-LR and CanESM5 demonstrate consistent performance across different regions and thus should be used as Thailand's preferred models. The implementation of CAMS-CSM1-0 alongside other models proves beneficial for localized prediction needs.

3.3. Daily precipitation projections and extreme events analysis

This study derived daily precipitation projections of Thailand under three SSP scenarios (SSP-345, SSP-370, SSP-585) for two CMIP6 climate models, MPI-ESM1-2-LR and CanESM5. In these scenarios, the three future time slices were analyzed: the near future (2025–2050), mid-century (2051–2075), and the end of the century (2076–2099). The selections of these models and SSPs are from a broad spectrum of GHG concentration trajectories, which are used to evaluate the impacts of these models against possible low, intermediate, and high emissions futures.

Since Thailand's climate is strongly influenced by its monsoon, characterized by significant seasonal variability between the wet (June–October) and dry (November–May) seasons, predictive models must account for this temporal heterogeneity. To tackle this, the modeling framework had a unified architecture for year-round precipitation prediction but explicitly included seasonal information as input features. In particular, the three major climatological seasons were represented using categorical encodings (summer, rainy, and winter) (Waqas, Humphries, Hlaing, & Ahmad, 2024). This approach internalized season-specific precipitation dynamics, including shifts in rainfall intensity, frequency, and duration, during monsoonal variability using DL models such as CNN-LSTM and DyNN-MEM.

From a scientific perspective, the seasonal indicators are embedded into the model architecture for two core purposes. It preserves the faithfulness of seasonal rainfall patterns by ensuring that the temporal pattern is not homogenized across the annual cycle. Second, it permits the seasonal precipitation signals to interact nonlinearly with atmospheric drivers, which is particularly crucial for regions such as Thailand, where seasonal precipitation is shaped by complex atmospheric processes, including the Indian monsoon, shifts in the Intertropical Convergence Zone (ITCZ), and regional convective activity.

The model's performance was assessed on a seasonal basis to validate its ability to reproduce both normal and extreme precipitation regimes in all seasons (Figure 7). Results from both CNN-LSTM and DyNN-MEM models show that they well captured the intra-seasonal variability. Still, their rainfall peaks were predicted during the monsoon season, especially for high-emission SSP585 scenarios. In the northeastern and southern regions (during the rainy season), precipitation intensities were increased significantly under these cases, as anticipated under increased monsoonal circulation and higher moisture convergence under global climate model projections.

While the CNN-LSTM model demonstrated a strong ability to capture high-intensity precipitation events, the DyNN-MEM (an ensemble-based model), on the other hand, provided more conservative estimates. This behavior is consistent with the observed behavior of ensemble modeling approaches, which often smooth out variability and, therefore, are less likely to make false optimistic predictions about extremes. However, both models exhibited high sensitivity to temporal shifts in seasonal precipitation. They

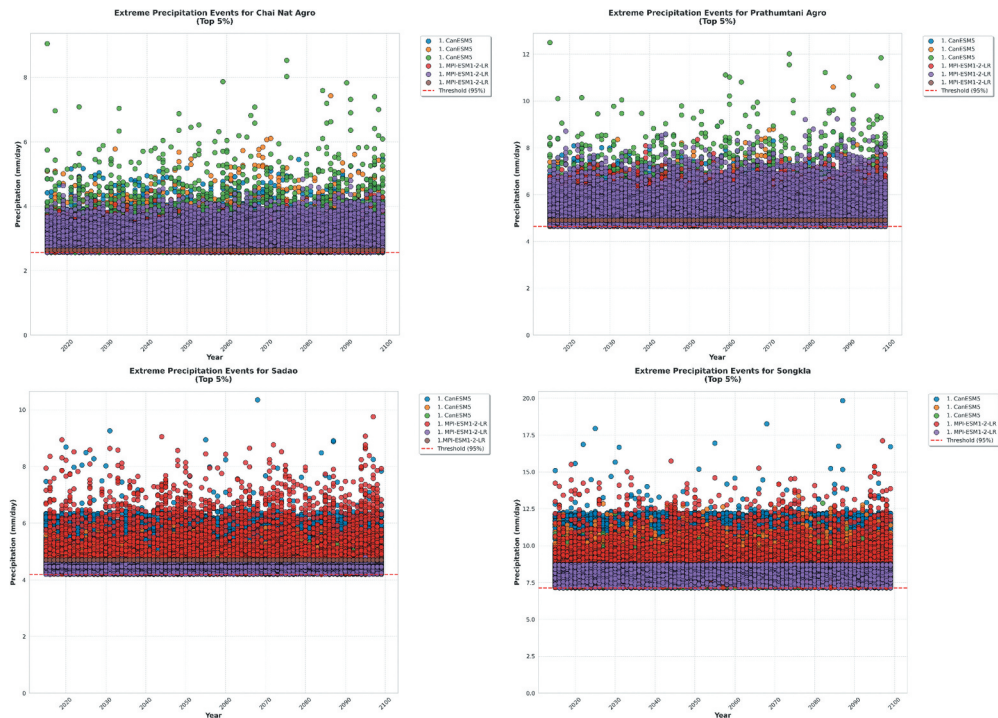


Figure 7. Extreme events (top %) for different stations under MPI-ESM1-2-LR and Can ESM5.

demonstrated the impact of increased rainfall in the southern coastal zones, which is particularly relevant for future studies in hydrological and agricultural planning activities.

A percentile-based thresholding technique was applied to quantify extreme rainfall behavior. Since the precipitation at each station is locally climatological, this study defined extreme precipitation events by those quantities that exceeded the 95th percentile of daily rainfall for each station. This method provides a robust statistical framework separating spatial heterogeneity in baseline precipitation from locations of extreme precipitation without imposing a strict limit on the maximum threshold of the recurrence interval.

The results show that the MPI-ESM1-2-LR model consistently exhibits higher frequencies and intensities of extreme events compared to CanESM5. Mean extreme precipitation for MPI-ESM1-2-LR projections, using Tak Pha Agro, Chai Nat Agro, and Prathumtani Agro stations as examples, ranges between 2.63 mm/day and 4.91 mm/day. On the other hand, the CanESM5 model predicts zero extreme events at some inland locations, with extremes ranging from being too small (systemic underprediction) to compromised model sensitivity to high-magnitude rainfall events.

There is a scientific reason for this discrepancy as it captures intermodel differences in portraying mesoscale convective processes, which dominate in impacts of short-duration and high-intensity rainfall in the tropical environment. Such convective dynamics are crucial for understanding localized flooding risks, and the MPI-ESM1-2-LR model may better represent resilient infrastructure design. These differences further suggest that multi-model ensemble assessments are needed, at least to inform an evaluation of future climate extremes, since projecting underestimates of the risk landscape can be achieved by relying on a single model.

Extreme rainfall was also spatially heterogeneous. Maritime influences, such as sea breeze convergence and increased latent heat availability, may explain the elevation of coastal stations' precipitation thresholds and maximum per station, as seen in Sadao (4.2 mm/day) and Songkhla (7.4 mm/day). The lower thresholds at inland stations, such as Nakhon Sawan and Lopburi, align with their more continental climatic regimes, characterized by reduced humidity and convective inhibition during the dry season.

The hotspots of extreme precipitation modeled (e.g., southern coasts and northeastern plains) are mechanically connected to the climatic-geography interactions in Thailand. The increase of extreme precipitation over the southern peninsula (e.g., Songkhla: 7.4 mm/day) might be related to the intensification of monsoonal moisture advection and sea-surface warming in SSP585, which increases convective available potential energy (CAPE). Conversely, the extremes of the northeastern portion could be a sign of increased mesoscale convective systems, which are more likely to be supported by the Mekong moisture plume. Their low biases in inland stations (e.g., Nakhon Sawan) may be evidence of the smoothing of localized convective processes in DyNN-Mem, and the high extremes of Hybrid LSTM-CNN may be overemphasizing unresolved topography-GCM interactions. These results demonstrate the importance of the ensemble approach in considering uncertainties in the description of land-atmosphere feedback.

The daily precipitation patterns from MPI-ESM1-2-LR (Hybrid CNN-LSTM) projections show uniform spatial distribution throughout different SSPs and periods (Figure 8). The precipitation projection under the SSP245 (2025–2050) shows a distribution from 0.0 to 20.0 mm, with higher precipitation in Thailand's northeastern and southern areas. The

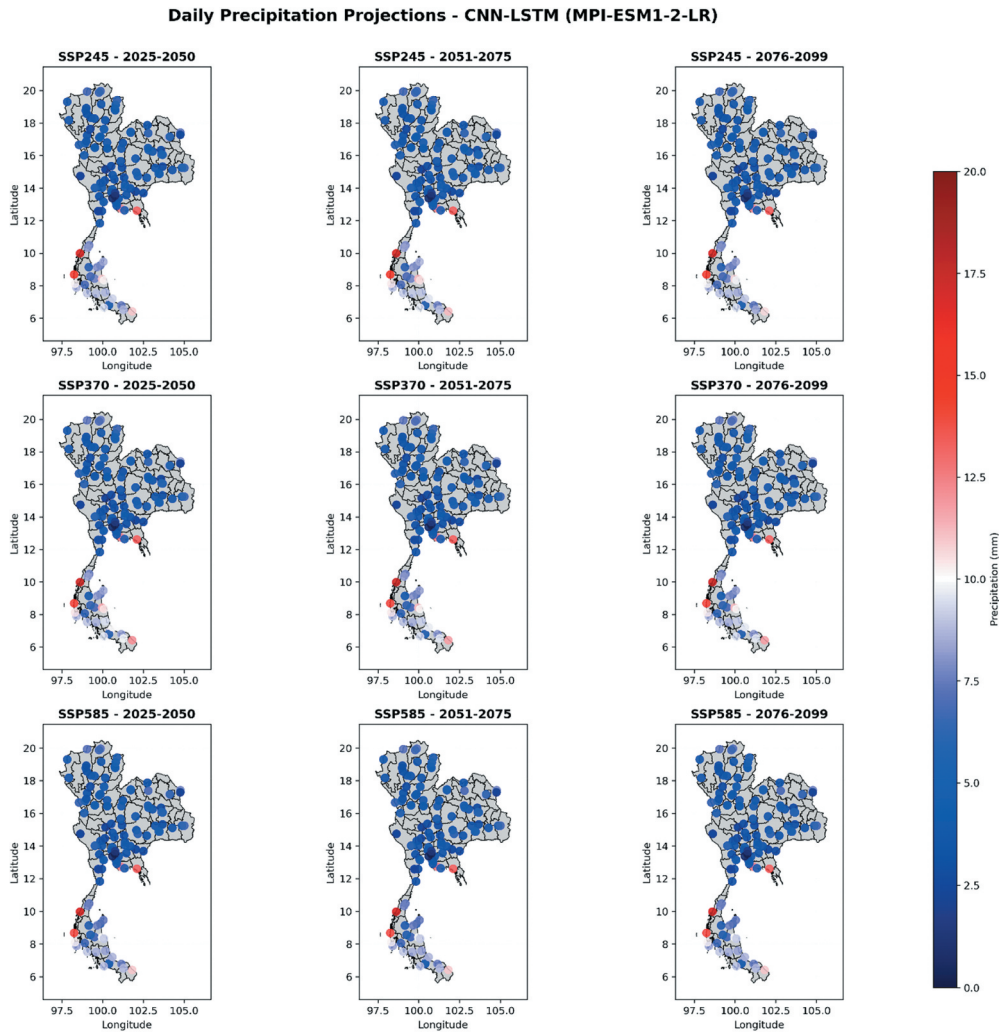


Figure 8. Precipitation projection with Hybrid LSTM-CNN (MPI-ESM1-2-LR).

SSP370 and SSP585 scenarios show increased precipitation intensity compared to the SSP245 while maintaining the same spatial pattern across all three scenarios in the high-emission SSP585 scenario. The projected precipitation intensity for 2051–2075 and 2076–2099 matches the initial spatial patterns but shows increased precipitation in regions under the SSP585 scenario, which reaches 20.0 mm. Higher emission scenarios indicate Thailand will face more severe rainfall events, which will impact the northeastern and southern regions of the country. The DyNN-MEM projections for MPI-ESM1-2-LR present daily precipitation between 0.0 and 7.5 mm during the same SSPs and periods (Figure 9).

The spatial distribution from DyNN-MEM projections matches CNN-LSTM results but indicates slightly lower precipitation levels. The discrepancy between DyNN-MEM and the other methods arises from the different approaches, as DyNN-MEM employs an ensemble methodology, resulting in more conservative estimates. The SSP245 scenario predicts a

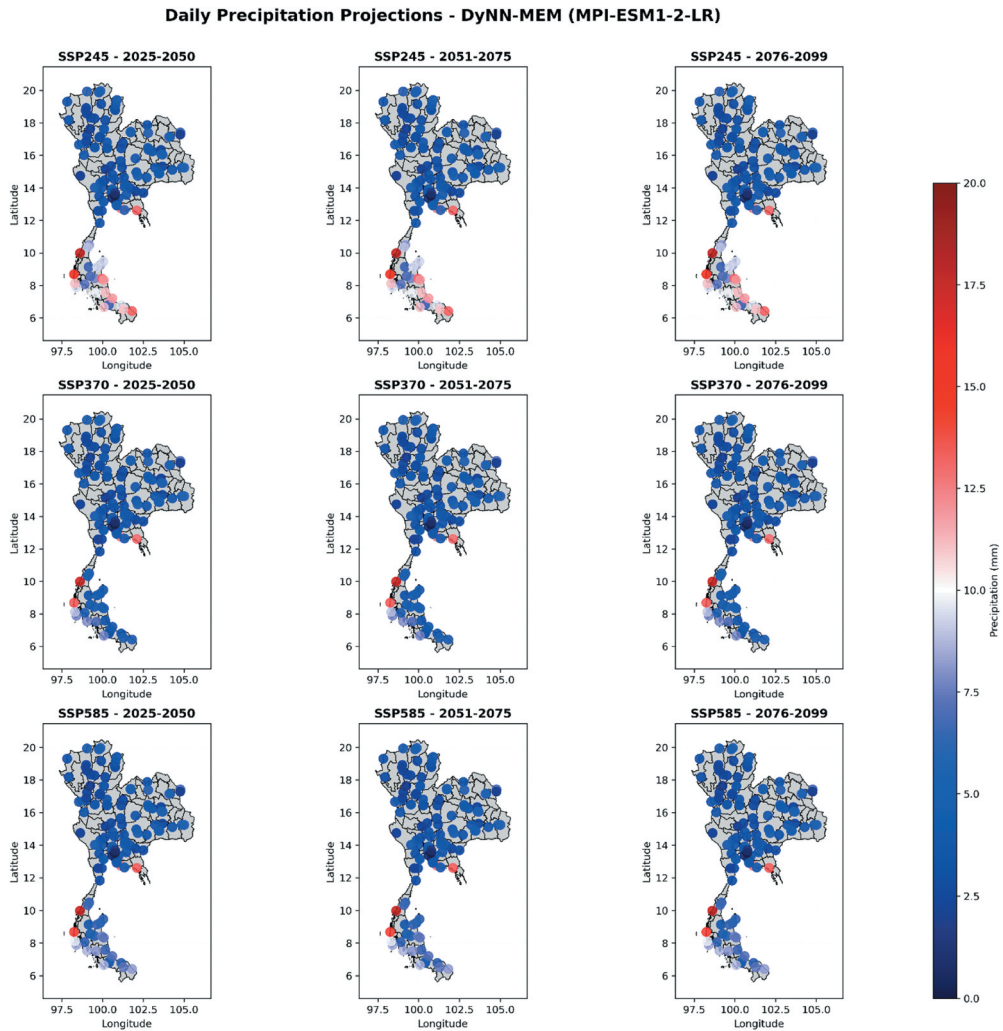


Figure 9. Precipitation projection with DyNN-Mem (MPI-ESM1-2-LR).

consistent precipitation pattern across 2076–2099, maintaining precipitation levels between 2.5 and 7.5 mm. The SSP585 scenario shows elevated precipitation intensities, reaching 7.5 mm in southern regions compared to the same period. According to research, higher emission scenarios suggest that Thailand will experience more intense precipitation events.

The CanESM5 model exhibits precipitation levels ranging from 0.0 to 20.0 mm in its CNN-LSTM projected spatial patterns across multiple SSPs and periods (Figures 10 and 11). The SSP245 scenario for 2025–2050 predicts moderate precipitation levels, with the highest accumulations occurring in northeastern and southern areas. The SSP370 and SSP585 scenarios from the CanESM5 model demonstrate rising precipitation intensity, with SSP585 reaching 20.0 mm in select regions. The analysis of precipitation intensity for the periods 2051–2075 and 2076–2099 reveals an ongoing trend of increasing precipitation levels over time. Higher emission scenarios indicate

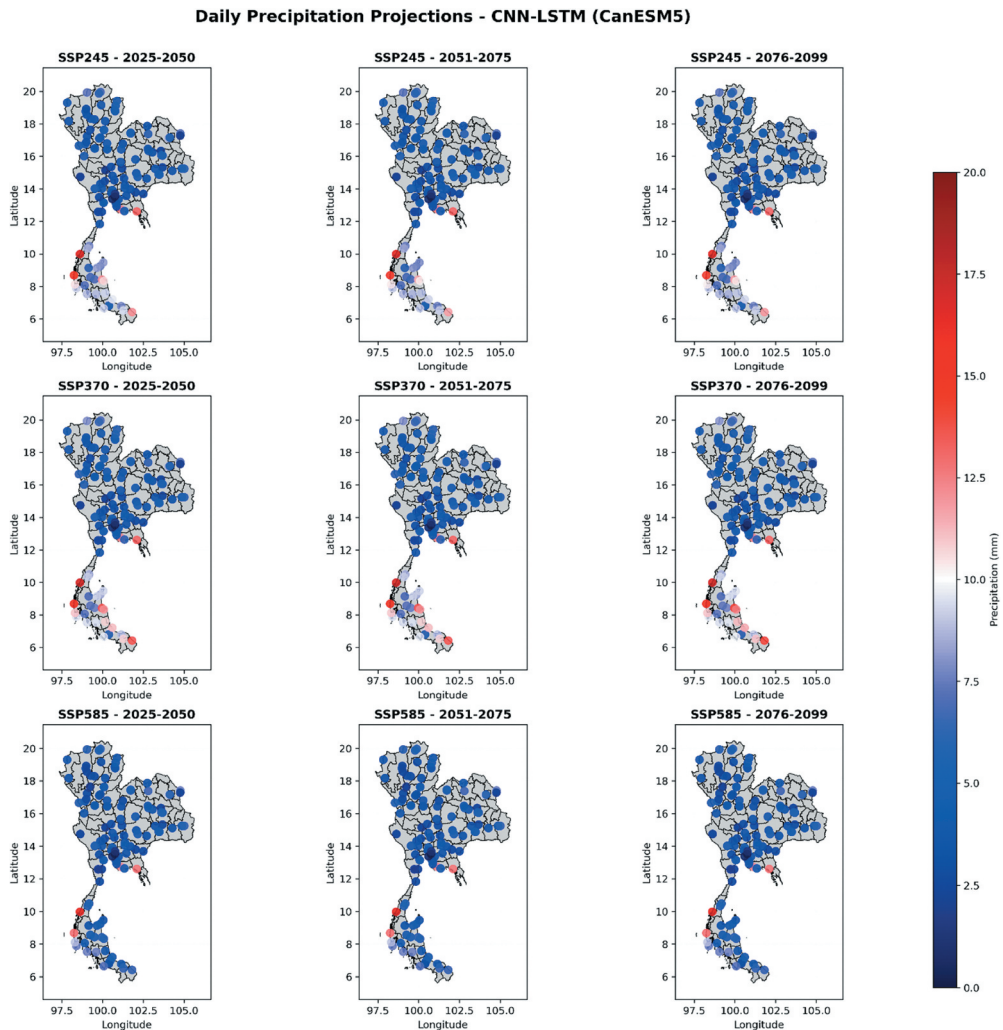


Figure 10. Precipitation projection with Hybrid LSTM-CNN (CanESM5).

Thailand will face increased frequency and intensity of rainfall, primarily in the northeastern and southern regions. The CanESM5 model in DyNN-MEM projection shows precipitation amounts between 0.0 and 7.5 mm. The spatial distribution from DyNN-MEM matches the CNN-LSTM projections; however, this method shows lower precipitation intensity within the SSP245 scenario for 2076–2099, ranging from 2.5 to 7.5 mm. The SSP585 scenario shows elevated precipitation intensities exceeding 7.5 mm across southern areas during the 2076–2099. The climate projection models MPI-ESM1-2-LR and CanESM5 indicate that Thailand will experience changes in precipitation patterns across various SSPs and time horizons. The northeastern and southern regions will experience greater precipitation intensities under high-emission SSP585 scenarios. The CNN-LSTM model detects elevated precipitation intensities, but DyNN-MEM generates more cautious estimates. The results demonstrate the need for researchers to evaluate future climate impacts using various modeling

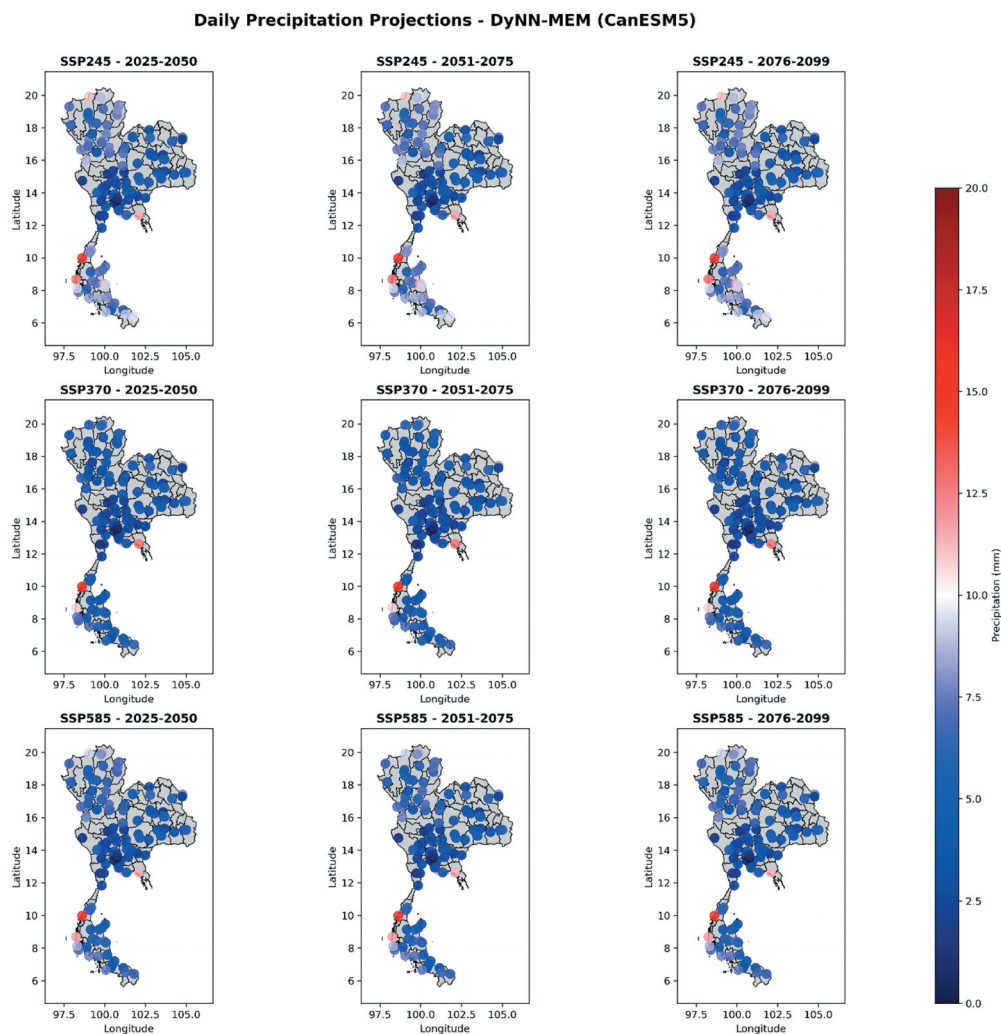


Figure 11. Precipitation projection with DyNN-Mem (CanESM5).

methods and scenario conditions. Higher precipitation intensities predicted under high-emission scenarios are expected to impact Thailand's regional hydrology, water resource management, and agricultural systems. The projections can guide policy-makers and stakeholders when they create adaptation and mitigation strategies to address climate change impacts.

4. Discussion

This study confirms previous research on AI-based models used for precipitation down-scaling and climate projections while providing new insights into the field. The DyNN-Mem model demonstrates exceptional performance in processing MPI-ESM1-2-LR and CAMS-CSM1-0 data, which supports recent findings on the ability of hybrid deep learning systems to detect complex patterns in climate data. Wang and Tian (2022) demonstrated

that DL-based models, such as the SRDRN, surpass traditional statistical methods, including QM, in preserving spatial and inter-variable relationships in downscaled precipitation data (Wang & Tian, 2022). Baño-Medina et al. (2020) demonstrated that CNNs paired with long LSTM produce more accurate downscaled precipitation data through their ability to recognize spatial patterns and temporal patterns simultaneously (Baño-Medina et al., 2020). The DyNN-Mem model combines MEM networks with attention mechanisms to enhance its capability for detecting long-term dependencies, as research has shown that such mechanisms deliver better climate projection results Santoro et al. (2016) and Zhang et al. (2021).

The DyNN-Mem model demonstrates robustness in processing precipitation data non-linearities and spatial correlations, as evidenced by consistent R^2 ranging from 0.55 to 0.78 and MAE ranging from 0.22 to 0.29 across different GCMs. The complex relationship between monsoons, topography, and climate variability in Thailand presents significant challenges to precipitation modeling (Waqas, Humphries, Hlaing, & Ahmad, 2024). The model demonstrates successful CMIP6-GCM precipitation data downscaling capabilities, aligning with previous research that utilizes AI-based methods for regional climate modeling. Ahmed et al. (2021) proved how CNN-LSTM hybrid models excel at capturing the spatial-temporal characteristics of soil moisture, which closely correspond to precipitation patterns (Ahmed et al., 2021). Singh et al. (2023) demonstrated how self-supervised learning methods enhance the accuracy of climate data downscaling in areas with challenging terrain and limited observational records (Singh et al., 2023).

The Hybrid LSTM-CNN model demonstrated competitive performance in specific scenarios yet displayed significant variations between different GCMs, resulting in R^2 ranging from 0.36 to 0.77. The model demonstrates sensitivity to the individual characteristics of input GCM data, which explains its variable performance noted in studies that use hybrid DL architectures (Baño-Medina et al., 2020). The Hybrid LSTM-CNN model demonstrated challenges in detecting extreme precipitation events and non-linear patterns, as its RMSE ranged from 1.28 to 3.29. The study supports Maraun and Widmann (2018) by demonstrating that traditional statistical downscaling methods fail to accurately represent the non-stationary and spatially correlated characteristics of precipitation data (Maraun & Widmann, 2018). The DyNN-Mem model implements MAN and attention mechanisms from advanced AI techniques, which overcome these limitations by enabling better retention and utilization of historical information for enhanced predictive accuracy.

The CMIP6-GCM ranking analysis identified MPI-ESM1-2-LR and CanESM5 as the best models for precipitation downscaling applications in Thailand. Research studies have already evaluated CMIP6-GCM performance in Southeast Asia, which supports this discovery. Humphries et al. (2024) recognized MPI-ESM1-2-LR as a leading model for predicting precipitation in southern Thailand because it successfully reproduced observed precipitation patterns with minimal variability and high reliability (Humphries et al., 2024). The research by Prathom and Champrasert (2023) demonstrated that CanESM5 provides consistent and accurate precipitation predictions across various climatic settings, making it an ideal choice for regional climate modeling applications (Prathom & Champrasert, 2023). These models deliver robust performance because they effectively model precipitation patterns across regions with complex topography and monsoon-driven climate systems.

Future precipitation projections under different SSPs demonstrate how climate change will affect Thailand's hydrological systems. The high-emission scenario

SSP585 exhibits rising precipitation intensity, which aligns with global observations of extreme rainfall increases attributed to global warming (Papalexiou & Montanari, 2019). The projected precipitation distribution shows elevated intensities across northeastern and southern Thailand, which matches previous research on regional precipitation responses to climate change. Tangang et al. (2017) predicted that Southeast Asia will experience more intense precipitation during the monsoon season due to global warming strengthening hydrological cycle patterns (Tangang et al., 2017). The predicted increase in precipitation intensities in Thailand's northeastern and southern regions will create significant challenges for water resource management, agricultural production, and disaster prevention, necessitating robust adaptation measures.

DyNN-Mem generates more cautious projections of future precipitation when compared to Hybrid LSTM-CNN under high-emission conditions. The ensemble methodology of DyNN-Mem utilizes multiple realizations to minimize uncertainty and enhance the reliability of its projections. Wang and Tian (2024) recommended ensemble-based methods to minimize bias and enhance accuracy in downscaled climate data, which supports the approach used in this study (Wang & Tian, 2024).

These results placed the performance of the AI model within the context of the Thai climate system. The advantage of DyNN-Mem is that it can implicitly learn monsoonal non-stationarity (e.g., ENSO phase shifts) and topographic effects (e.g., windward lifting), which are key limitations of traditional methods. Geographic heterogeneity of extremes, however, which is linked to coastal convergence zones and high-altitude regions, necessitates the direct addition of geographic covariates (e.g., elevation and land-sea masks) in later model versions. Stakeholders can use these results to identify high-risk regions (e.g., southern coasts under SSP585) where adaptation strategies (e.g., flood-resilient infrastructure) will need to consider compound risks associated with intensifying convection and sea-level rise. The risk-averse conservative DyNN-Mem projections provide a risk-averse policy baseline, and the uncertainty of the Hybrid LSTM-CNN models highlights the need for multi-model frameworks in climate adaptation planning. These tools demonstrate excellent potential to enhance regional climate projection accuracy and support climate adaptation initiatives through their combined robust performance with DyNN-Mem. The research outcomes enhance knowledge of AI-based climate modeling while delivering essential information to Thai policymakers and stakeholders. The research reveals that DyNN-Mem emerges as the optimal AI model for Thai precipitation downscaling, and MPI-ESM1-2-LR and CanESM5 demonstrate the best performance as CMIP6-GCMs for regional climate modeling in Thailand.

5. Conclusion

This study developed and tested two advanced AI-based downscaling models, DyNN-Mem and Hybrid LSTM-CNN, to produce precipitation projections over Thailand using CMIP6-GCM. Statistical measures were used to evaluate the model's performance in terms of R^2 , MAE, RMSE, and Pearson correlation coefficient. The proposed DyNN-Mem outperformed the Hybrid LSTM-CNN model on various GCMs that incorporated convolutional, recurrent, and memory-augmented attention mechanisms. It obtained an R^2 range of 0.55 to 0.78, MAE in the range of 0.22 to 0.29, and RMSE in the range of 1.28 to 3.30, and was able to capture the spatiotemporal variance and extreme precipitation events.

Two selected GCMs (MPI-ESM1-2-LR and CanESM5) were found to be the most suitable for Thailand, with MPI-ESM1-2-LR showing outstanding and spatially robust predictive skills. The ensemble means of DyNN-Mem, and MPI-ESM1-2-LR offered the strongest and sturdiest structure of future precipitation projection over the area. The downscaled outputs have significant repercussions on sectoral planning and decision-making in real practice. The data has the potential to provide operational information for reservoir operations, watershed planning, and long-term climate adaptation policies at the national and regional levels of water resource management.

Despite the promising results of the research, it is constrained by the presence and quality of observational data as well as the computational demands of deeply parametrized learning models. Sparse station data and/or that which is not consistent over time may limit the model to simulate rare or extreme events.

Additional research should incorporate multi-source data, including satellite and reanalysis products, to enhance the generalizability and stability of the models. This framework can be generalized to other climate variables, such as temperature extremes, and the reliability of the projections may be further improved by ensemble learning with physical constraints. Moreover, the methodological framework developed in the present study has a significant likelihood of transfer to the rest of Southeast Asia and even to data-limited tropical regions with similar hydroclimatic concerns. The future development of AI-based downscaling systems will be particularly significant in supporting the estimation of region-specific climate impacts and the design of climate-resilient infrastructure.

Abbreviations

AI	Artificial Intelligence
CMIP6	Coupled Model Intercomparison Project Phase 6
GCM	Global Climate Model
DyNN-Mem	Dynamic Neural Network with Memory
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
TMD	Thai Meteorological Department
R^2	Coefficient of Determination
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
SSP	Shared Socioeconomic Pathway
IPCC	Intergovernmental Panel on Climate Change
RCM	Regional Climate Model
IDW-ANN	Inverse Distance Weighting-Artificial Neural Network
SRDRN	Super-Resolution Deep Residual Network
QM	Quantile Mapping
DL	Deep Learning
ML	Machine Learning
ELU	Exponential Linear Unit
ReLU	Rectified Linear Unit
MSE	Mean Squared Error
IQR	Interquartile Range

WCRP	World Climate Research Program
ESGF	Earth System Grid Federation
CAMS-CSM1-0	Chinese Academy of Meteorological Sciences Climate System Model version 1.0
MPI-ESM1-2-LR	Max Planck Institute Earth System Model version 1.2 Low Resolution
FGOALS-g3	Flexible Global Ocean-Atmosphere-Land System model version g3
CanESM5	Canadian Earth System Model version 5
CNRM-CM6-1	Centre National de Recherches Météorologiques Climate Model version 6.1
CNRM-ESM2-1	Centre National de Recherches Météorologiques Earth System Model version 2.1
IPSL-CM6A-LR	Institut Pierre-Simon Laplace Climate Model version 6A Low Resolution
CWD	Consecutive Wet Days
PDF	Probability Density Function
NASA	National Aeronautics and Space Administration
MODIS	Moderate Resolution Imaging Spectroradiometer
RNN	Recurrent Neural Network
GAN	Generative Adversarial Network
QDM	Quantile Delta Mapping
CLIMEA-BCUD	Climate Model Evaluation and Bias Correction for Urban Development

Disclosure statement

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Data availability statement

The data utilized in this study is available upon request.

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